Pyramids of Number & Biomass

Exam code: 0610 & 0970

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Written by: Lára Marie McIvor
Reviewed by: Lucy Kirkham
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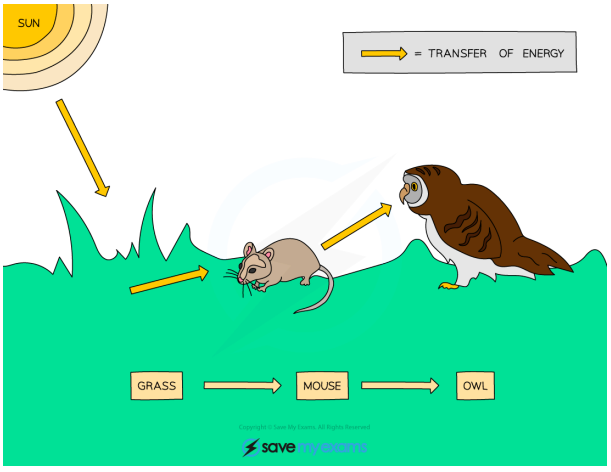
Help me understand

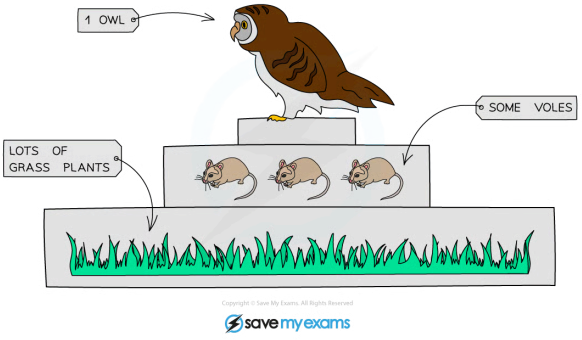
Pyramids of Numbers

- A pyramid of numbers shows **how many organisms** are present at each level of a food chain.
  - The **width of the box** indicates the **number of organisms** at that trophic level
- For example, consider the following food chain:



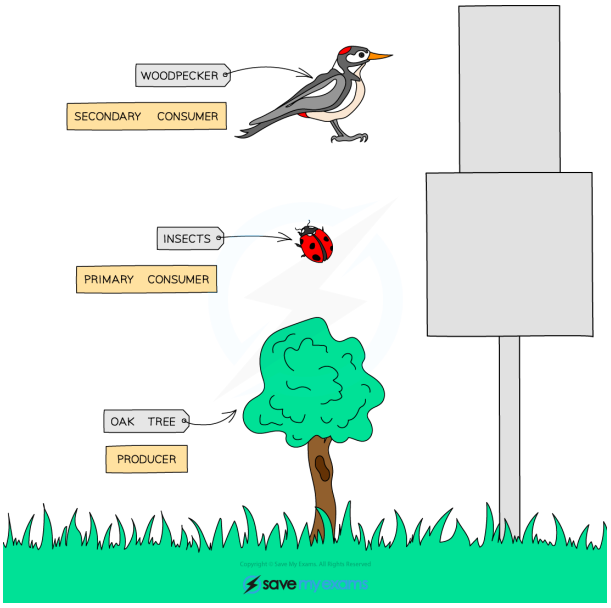
A food chain shows the transfer of energy

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



A pyramid of numbers

- A pyramid of numbers **doesn't always have to be pyramid-shaped**; this is because the **size of the organism is also important**
  - One large organism, like the oak tree in the pyramid below, contains enough energy to support many smaller organisms



Pyramids of numbers are not always pyramid-shaped

Rules to remember when drawing a pyramid of numbers

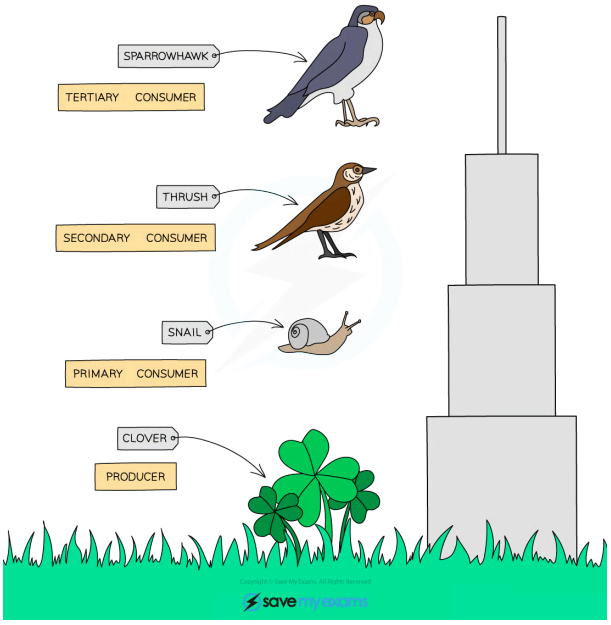
- You **cannot change the trophic level of the organisms** - they must stay in the same order as in the food chain:
  - producers on the bottom,
  - followed by primary consumers,
  - then secondary consumers,
  - then tertiary consumers

- Each bar should be **labelled** with its trophic level
- If graph paper is provided then the pyramid should be **drawn to scale**

Pyramids of Biomass

- A pyramid of biomass shows **how much living matter** is present at each trophic level of a food chain
  - Biomass is often a measure of the dry mass of an organism, once the water has been removed from the tissues

- Pyramids of biomass are **always pyramid-shaped**, regardless of what the pyramid of numbers for that food chain looks like
- This is because the **mass of organisms decreases as you go up a food chain**



A pyramid of biomass

- It can be advantageous to use a pyramid of biomass rather than a pyramid of numbers because:
  - pyramids of biomass **take into account the size of organisms** at each trophic level, so provide a more accurate representation of the living matter within an ecosystem
  - energy is stored in biomass, so biomass can be used as an **approximate representation of the energy available** at each level of a food chain

Examiner Tips and Tricks

Pyramids of number can be any shape, so make sure you learn the rules for drawing a pyramid of numbers as it is more common to see unusual shapes in the exam.

but it is very unlikely you will see this in an exam.

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